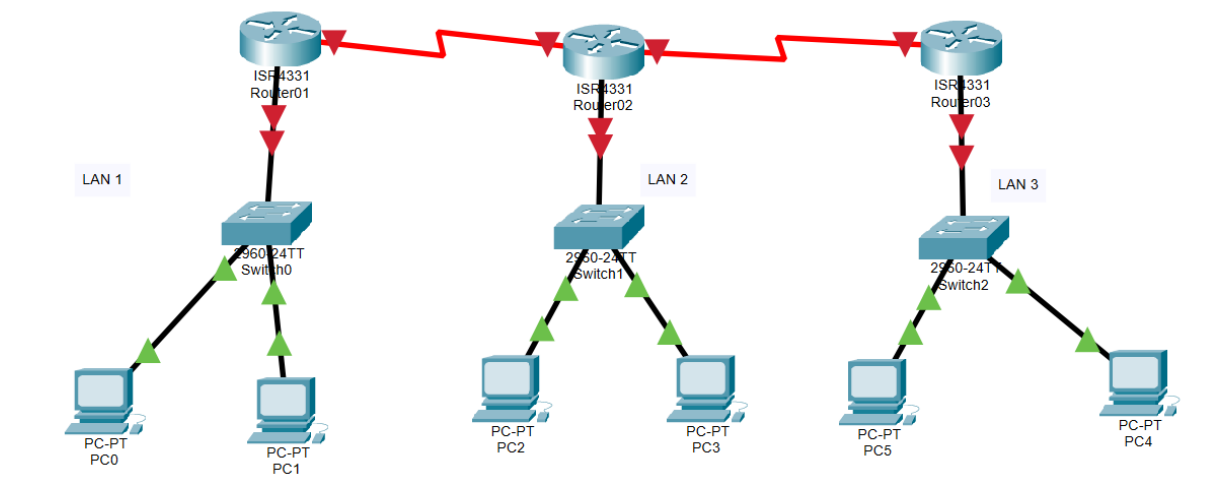


Subnetting

1. Consider a large enterprise that requires at least 3 subnets (LAN1, LAN2 & LAN3) and has chosen the private address 192.168.1.0/24 as its internal network address.

Find the 3 subnets, Range of usable IP Addresses and the broadcast addresses



Solution 1

192.168.1.0/24

Network portion	Host
11000000.10101000.00000001	00000000

Subnet Mask: 255.255.255.0

$2^n = \text{\#subnets}$

N= number of bits borrowed (Host portion)

$2^2 = 4$ subnets

Jump value or Magic number= last borrowed bit = **64**

LAN 1

Network address	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address
192.168.1.0/26	255.255.255.192	192.168.1.1	192.168.1.62	192.168.1.63

LAN2

Network address	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address
192.168.1.64/26	255.255.255.192	192.168.1.65	192.168.1.126	192.168.1.127

LAN 3

Network address	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address
192.168.1.128	255.255.255.192	192.168.1.129	192.168.1.190	192.168.1.191

2. Network address: 192.168.10.0/24

Subnet mask: 255.255.255.0

Create 8 subnets

Solution 2

$2^n = \text{\#subnets}$

$2^3 = 8$

192.168.10.0 Network 1 or Network zero

Network portion	Host
11000000.10101000.00001010	00000000

New Subnet mask

255.255.255.224:/27

11111111.11111111.11111111.11100000

Jump value: 32

192.168.10.00100000

Net 1

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.0/27	255.255.255.224	192.168.10.1	192.168.10.30	192.168.10.31

Net 2

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.32	255.255.255.224	192.168.10.33	192.168.10.62	192.168.10.63

Net 3

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.64	255.255.255.224	192.168.10.65	192.168.10.94	192.168.10.95

Net 4

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.96	255.255.255.224	192.168.10.97	192.168.10.126	192.168.10.127

Net 5

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.128	255.255.255.224	192.168.10.129	192.168.10.158	192.168.10.159

Net 6

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.160	255.255.255.224	192.168.10.161	192.168.10.190	192.168.10.191

Net 7

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.192	255.255.255.224	192.168.10.193	192.168.10.222	192.168.10.223

Net 8

Network Address	Subnet Mask	First IP Address	Last IP Address	Broadcast Address
192.168.10.224	255.255.255.224	192.168.10.225	192.168.10.254	192.168.10.255